#Videolink = https://drive.google.com/file/d/1YJUPjBjL2i98hvCjzkXrMxLUeKRafIuI/view?usp=sharing

TASK 1 SUMMARY :-

#1. **Column Formatting:** It strips invisible trailing spaces from your headers. Looking at your raw data, columns like Personalized_Recommendation_Frequency actually appear twice, once with a space at the end. Stripping them ensures you can reference them easily.

#2. **Exact Duplicates:** drop_duplicates() scans the entire row and removes it if another row matches it exactly 1-to-1.

#3. **Transaction Duplicates:** Since your data includes a transaction column (e.g., 814116), the script assumes this is a unique ID. If a user accidentally submitted twice under the same ID, the script removes the duplicate and keeps the last entry.

#4. I**nconsistent Ages:** It converts the age column to numeric values. It then drops rows where the age is below 13 or above 100, removing fake or typed-in error responses.

#5. **Standardization:** It cleans up text formatting in the Gender column so that "female" and "Female" aren't counted as two separate categories.

#6. **Trailing Spaces Removed:** "Rating_Accuracy " is now successfully renamed to "Rating_Accuracy".

#7. **Data Types:** All targeted columns are now confirmed as int64 (integer) or float64, which allows you to perform mathematical operations like calculating the mean, median, or correlation.

```
In [1]: %runfile 'D:/Course Data DS/PROJECT 6 FINALE/TASK 1.py'
wdir
Original shape: (800, 24)
Shape after dropping exact duplicates: (800, 24)
Shape after dropping duplicate transactions: (798, 24)
Shape after filtering invalid ages: (675, 24)
Shape after dropping missing critical values: (675, 24)
Cleaned dataset saved as: Amazon_cleaned.csv
```

TASK 2 SUMMARY :-

#1. Age Summary

The customers in this dataset range from 3 to 67 years old, with an average age of approximately 35.17 years. The distribution shows a fairly even spread across various age groups.

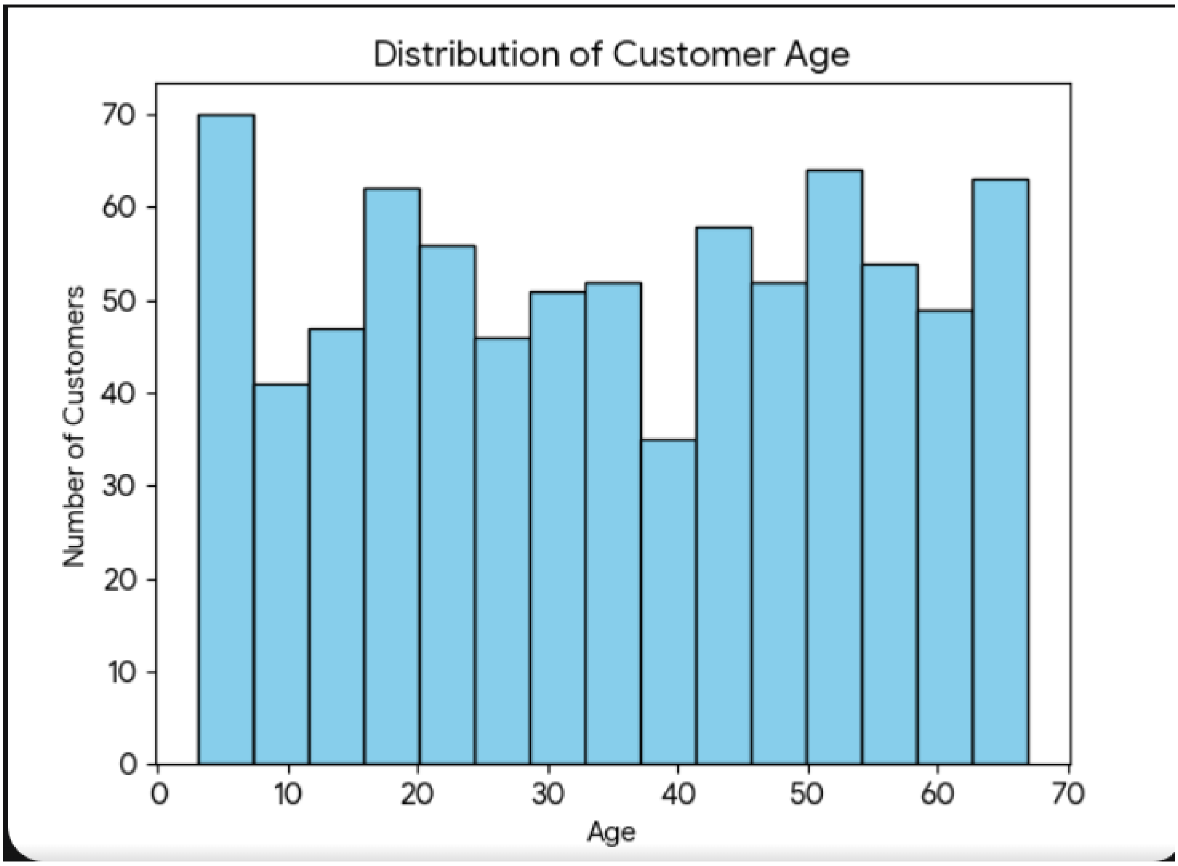
- **Mean Age:** 35.17
- **Median Age:** 35.5
- **Minimum Age:** 3
- **Maximum Age:** 67
- **Standard Deviation:** 18.92

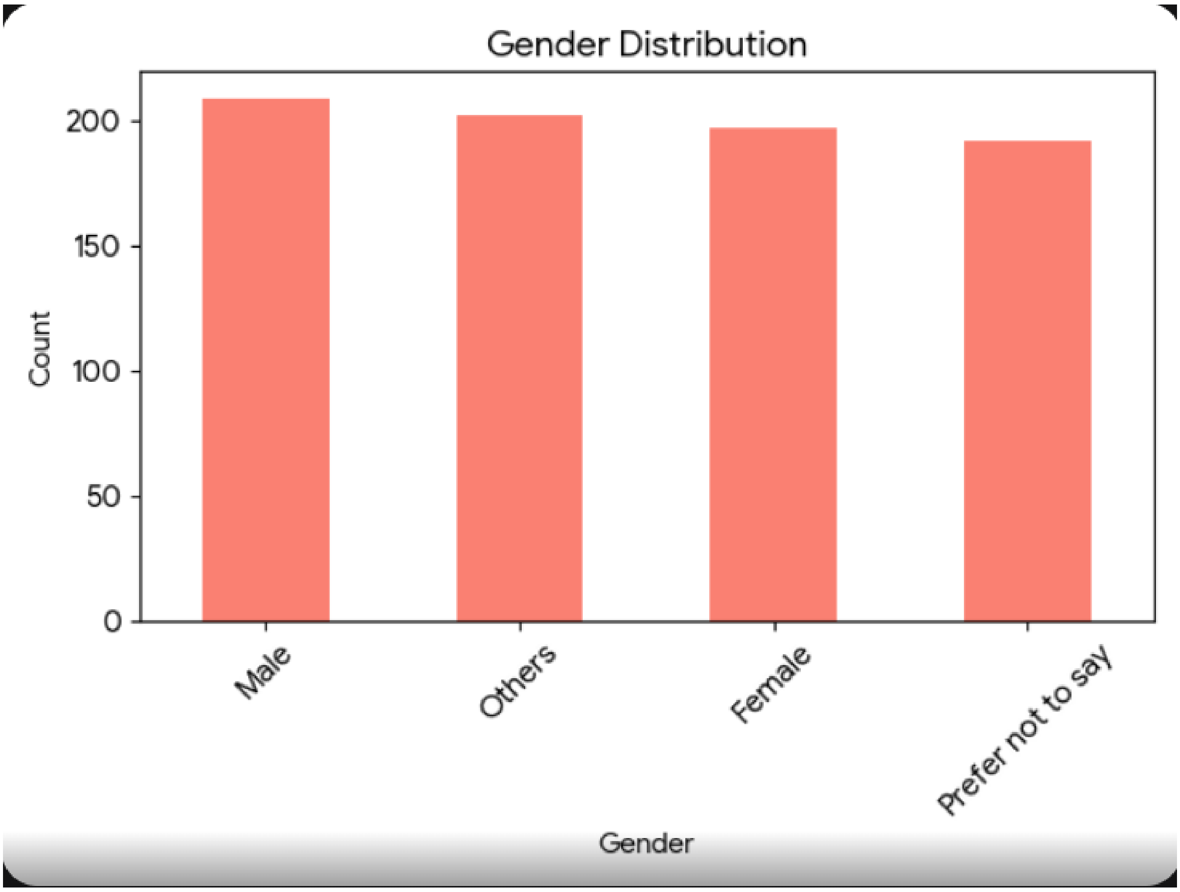
#2. Gender Distribution

The survey respondents are distributed almost equally across the four gender categories identified in the data:

Gender	Count
Male	209
Others	202
Female	197

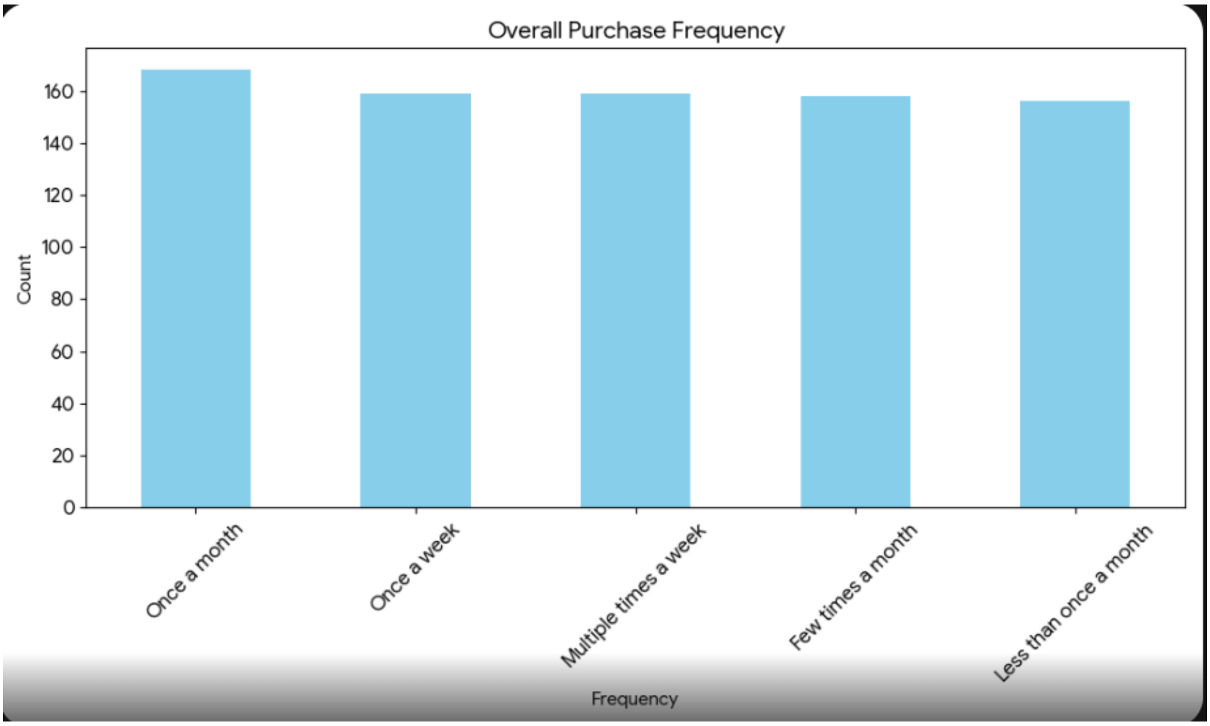
Gender	Count
Prefer not to say	192





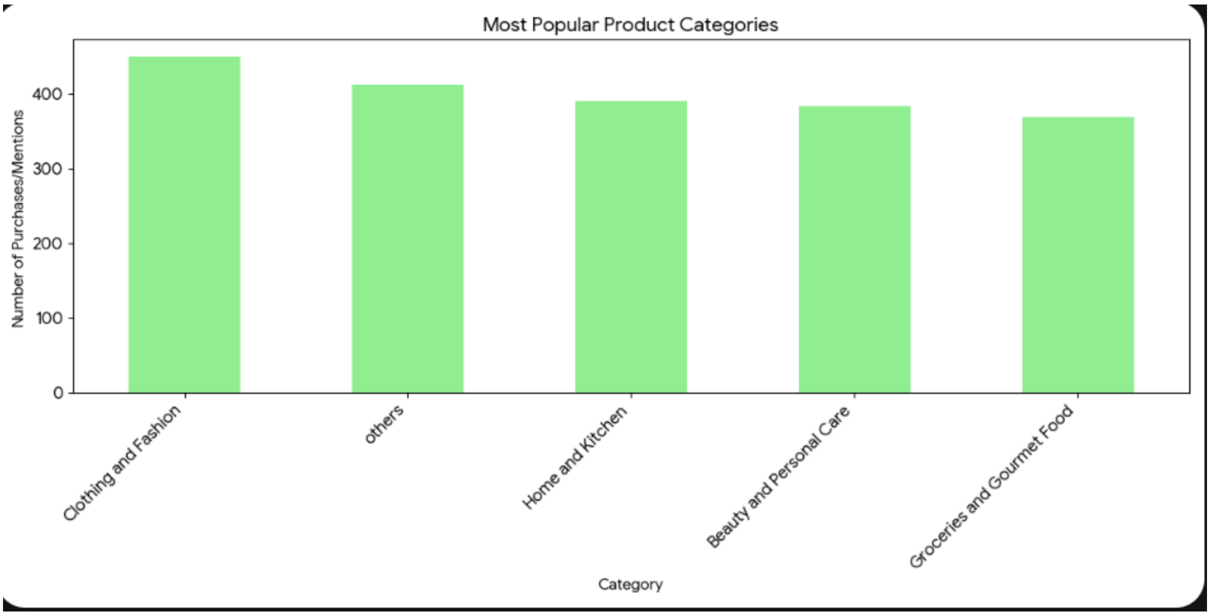
#3. **Purchase Frequency Analysis**

The data shows a very balanced distribution of shopping habits among respondents. Most customers shop at least once a month, with "Once a month" being the most frequent response, followed closely by more active weekly shoppers.



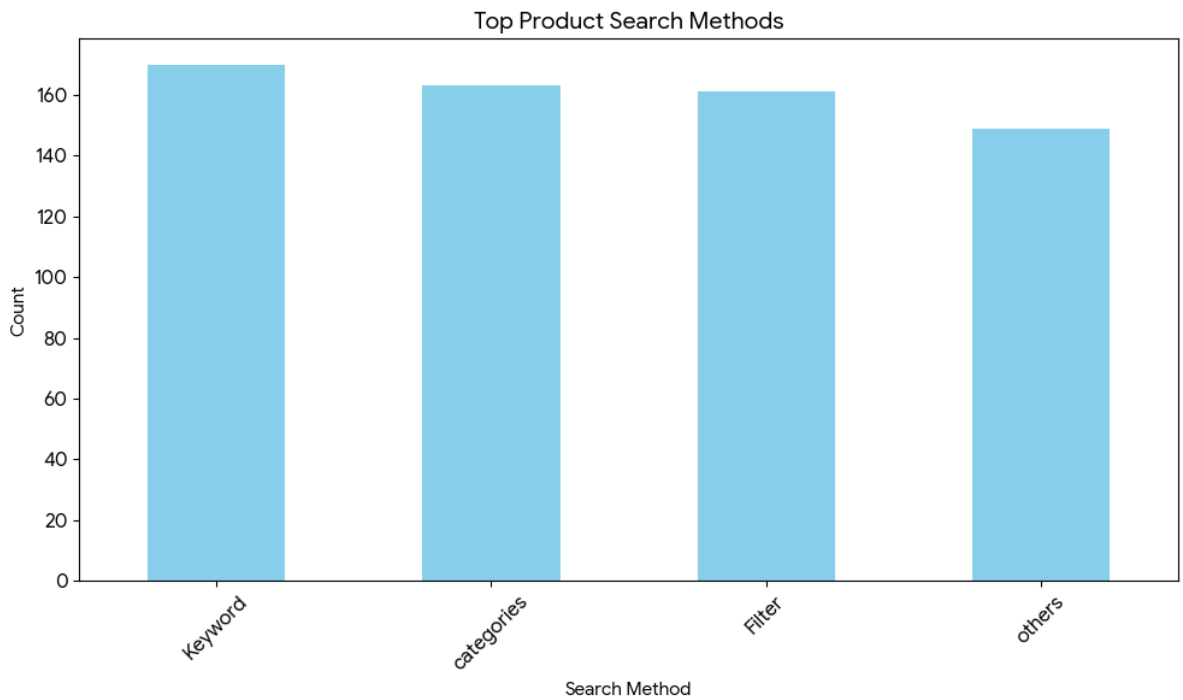
#4. **Popular Product Categories**

Since respondents could select multiple categories, I split the entries to count each category individually. **Clothing and Fashion** is the most popular category among the surveyed customers.



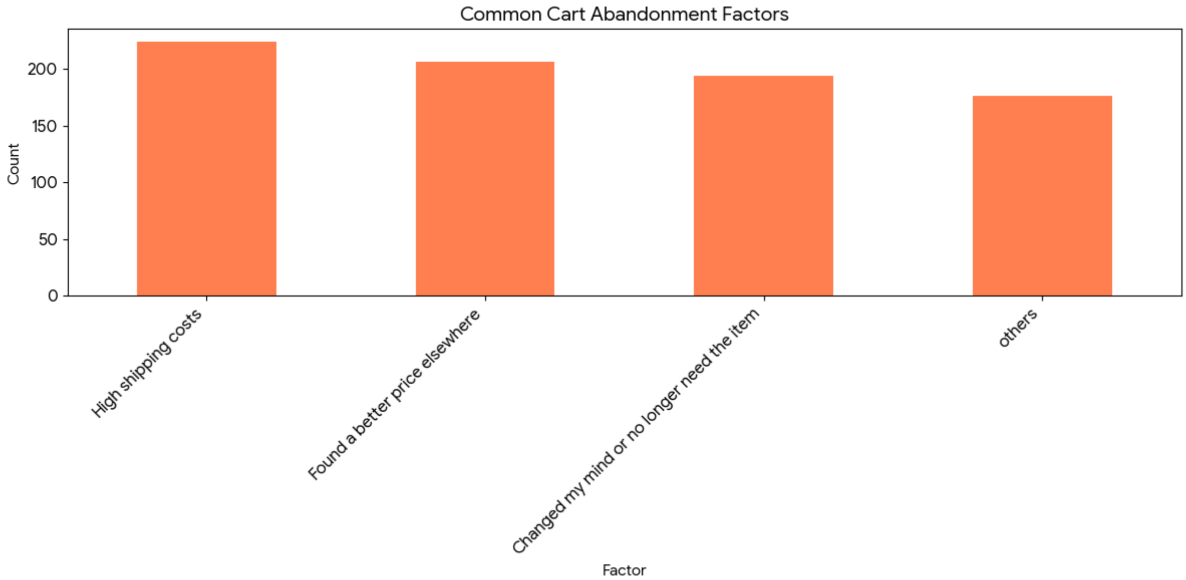
#5. **Top Product Search Methods**

The search methods used by customers are fairly evenly distributed, with **Keywords** being the most popular starting point for finding products on Amazon.



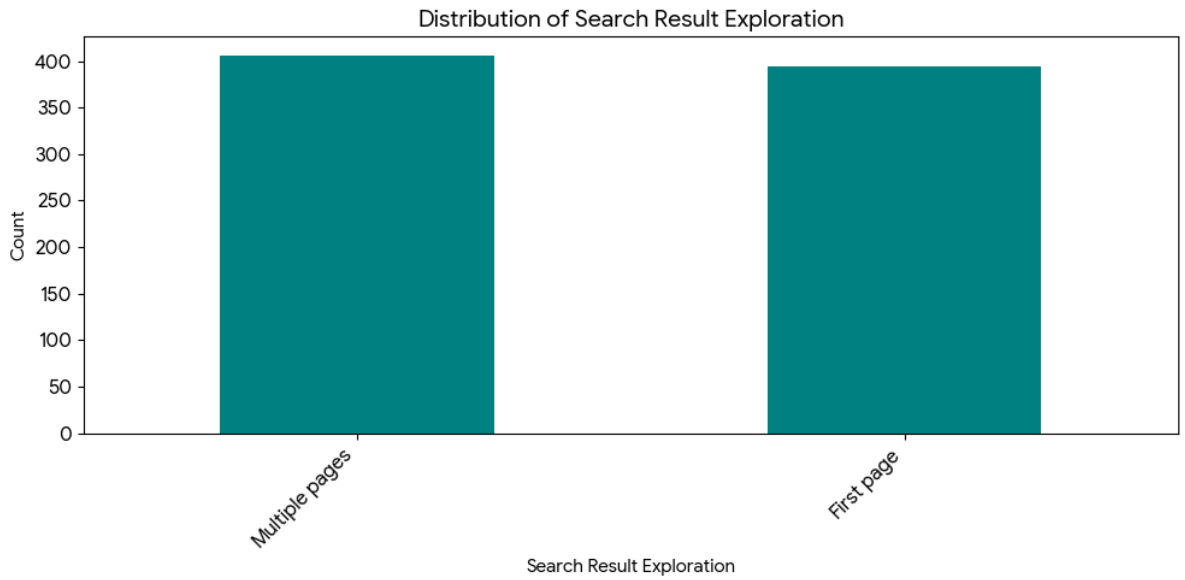
#6. **Common Cart Abandonment Factors**

Price and shipping logistics are the primary drivers for cart abandonment. **High shipping costs** was the most frequently cited reason why customers chose not to complete a purchase.

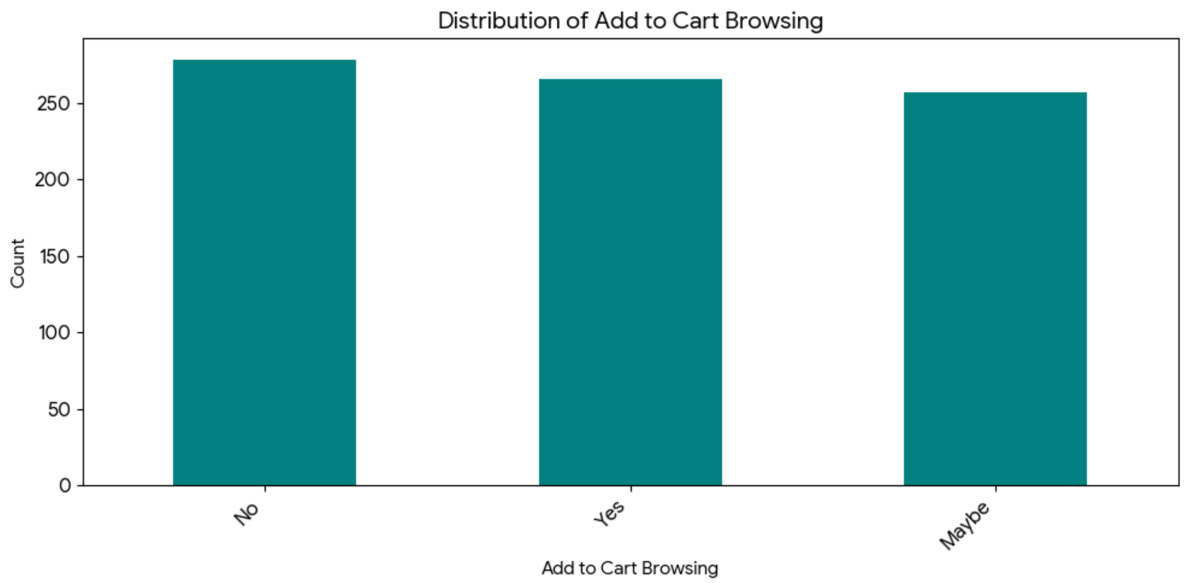


#7. **Key Behavioral Insights**

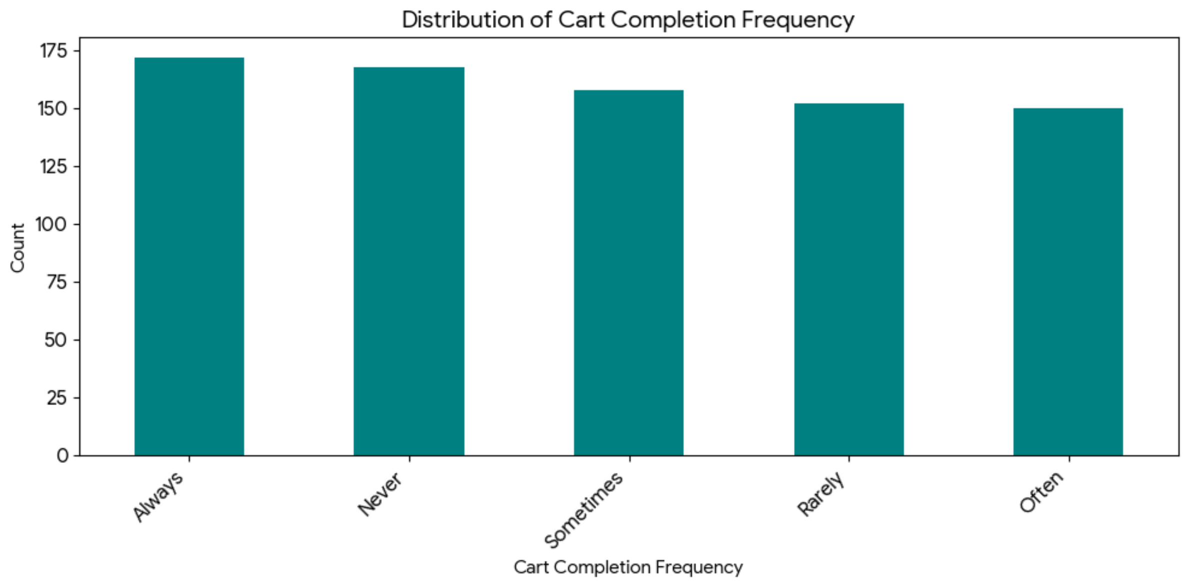
Behavioral Variable	**Most Frequent Response (Mode)**	**Count**
Purchase Frequency	Once a month	168
Browsing Frequency	Rarely	231
Search Exploration	Multiple pages	406
Add to Cart Browsing	No	278

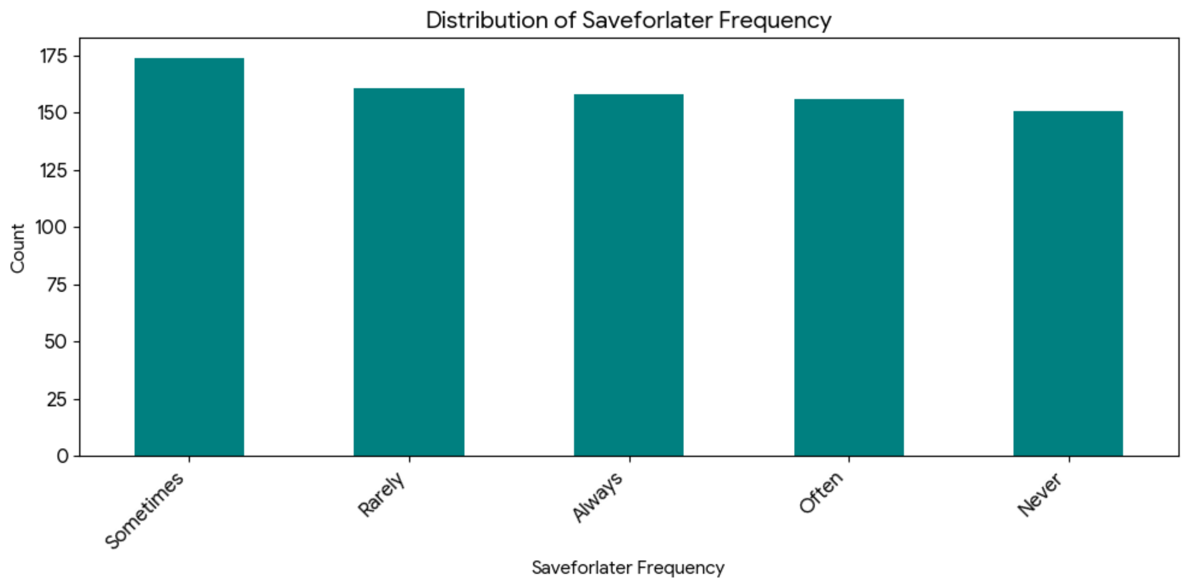


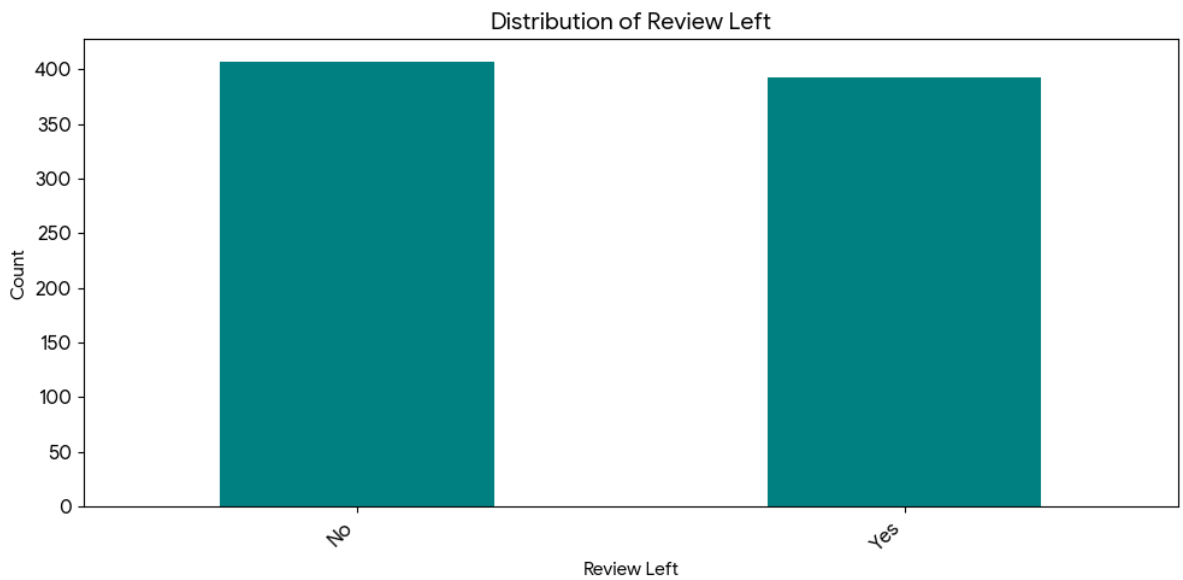
Distribution of Search Result Exploration
Count
0
50
100
150
200
250
300
350
400
Multiple pages
First page
Search Result Exploration



Distribution of Add to Cart Browsing
Count
0
50
100
150
200
250
No
Yes
Maybe
Add to Cart Browsing

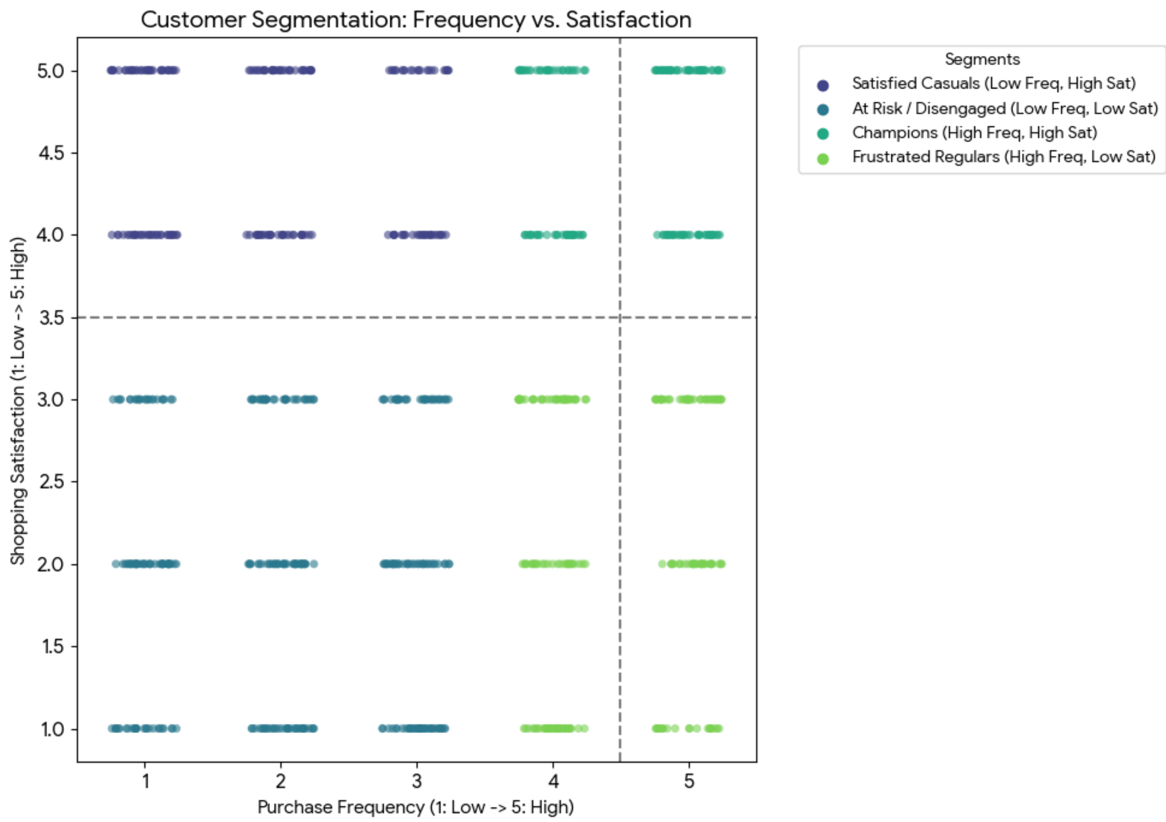

Distribution of Cart Completion Frequency
Count
175
150
125
100
75
50
25
0
Always
Never
Sometimes
Rarely
Often
Cart Completion Frequency


Distribution of Saveforlater Frequency
Count
175
150
125
100
75
50
25
0
Sometimes
Rarely
Always
Often
Never
Saveforlater Frequency


Distribution of Review Left
Count
400
350
300
250
200
150
100
50
0
No
Yes
Review Left

TASK 3 SUMMARY :-

#1. To perform this segmentation, I mapped the "Purchase Frequency" labels to a numerical scale (1-5) and used a 2 \times 2 matrix split at the midpoint of 3.5.



#2. **Demographic Differences (Age and Gender)**

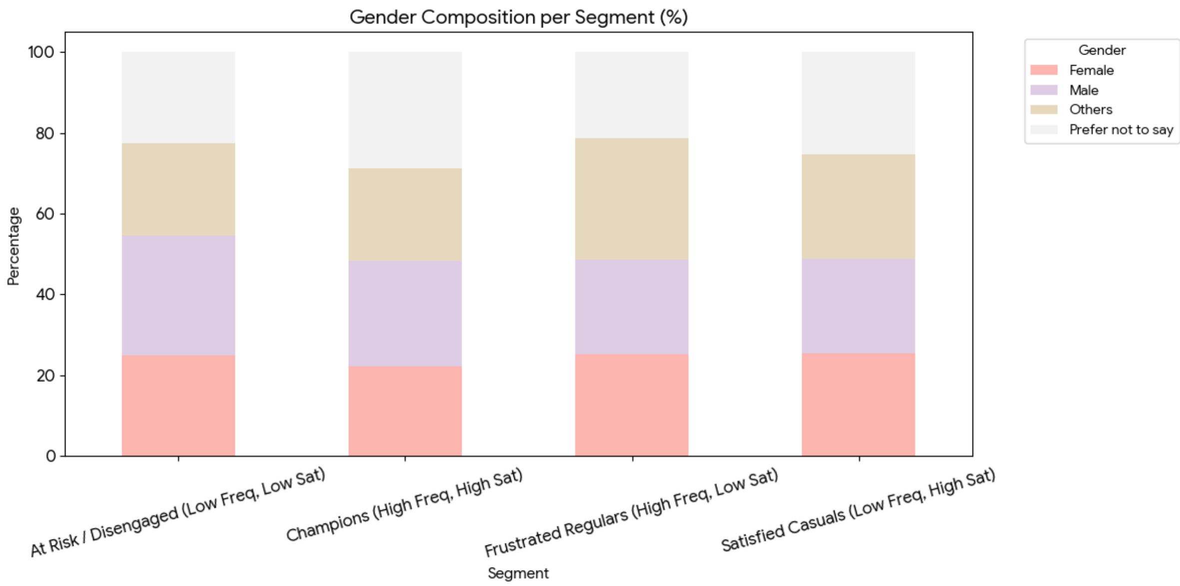
The age and gender distribution remain relatively balanced across all segments, indicating that shopping satisfaction and frequency on the platform are not tied strictly to a single age bracket or gender identity.

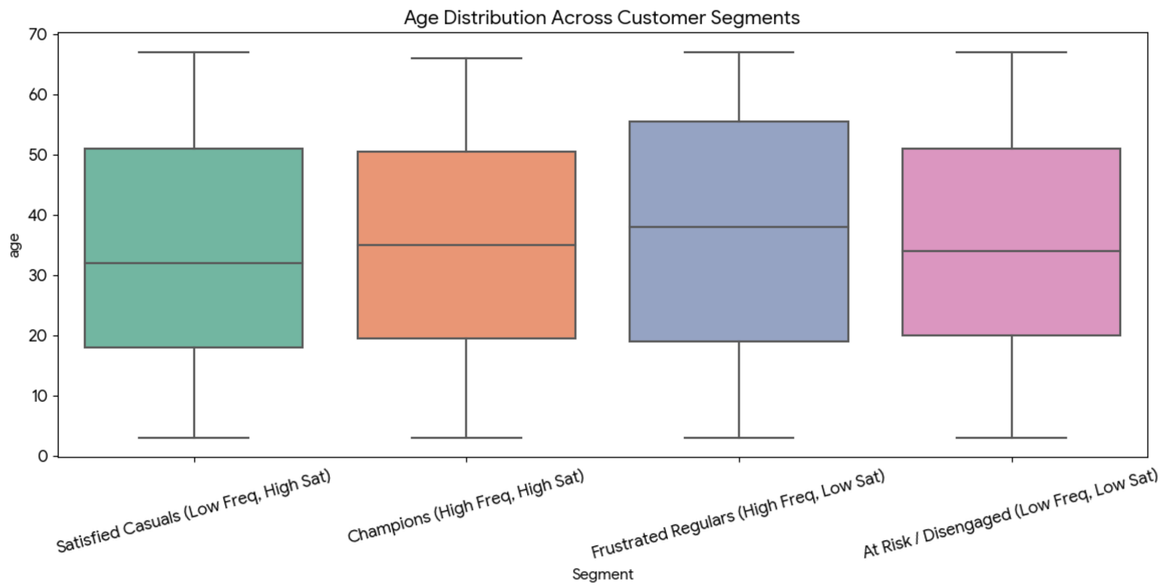
#3. **Behavioral Differences (Browsing Frequency)**

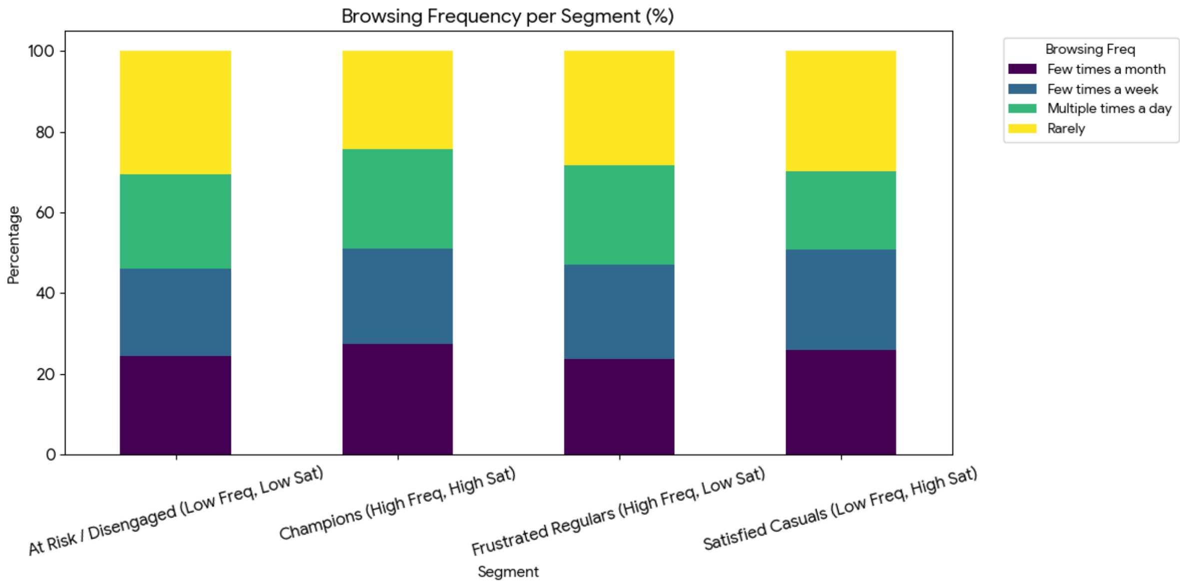
Browsing patterns reveal how each segment uses the platform before making a purchase decision.

#4. **Cart Abandonment Drivers by Segment**

Understanding why each segment drops off at the final step is crucial for targeted marketing.

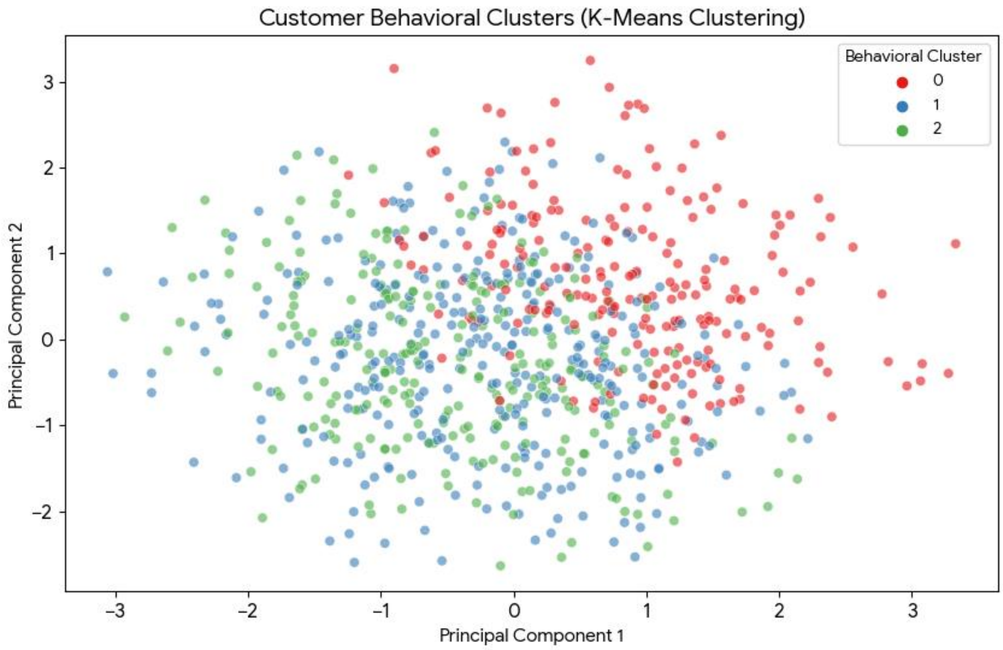






#5. K-Means Clustering on the behavioral survey responses to group customers with similar platform usage and decision-making habits.

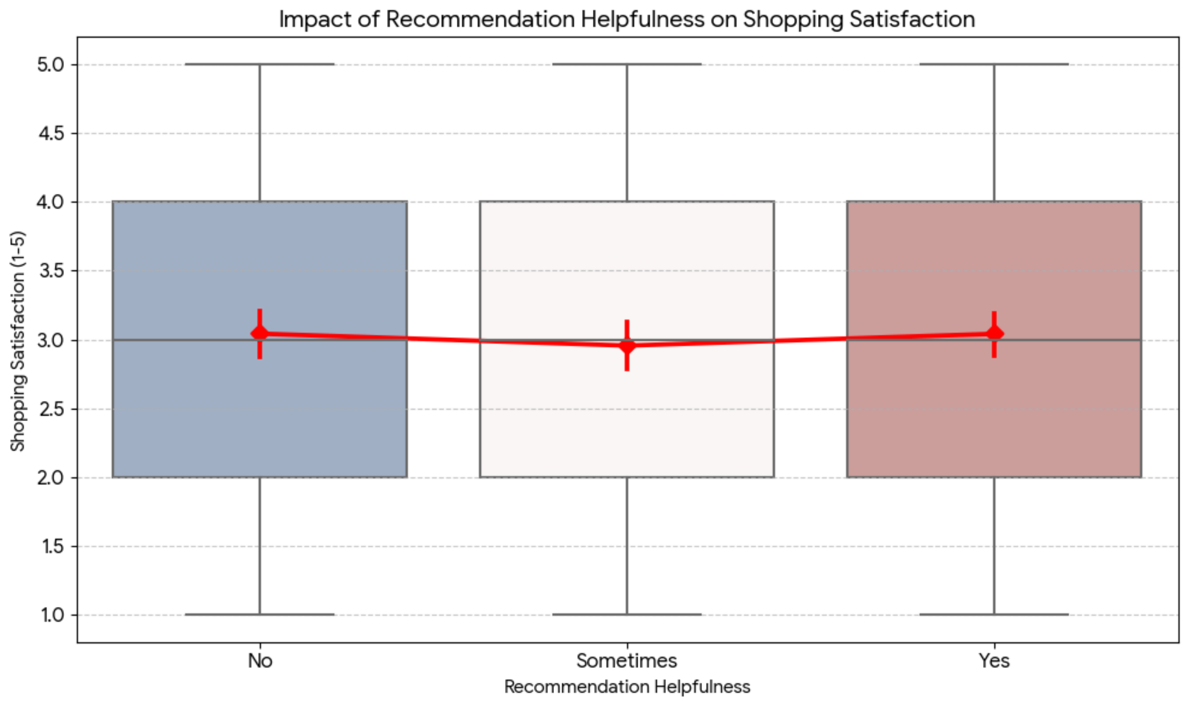
To ensure the clustering process was effective, Its mapped all qualitative responses (example - *Always, Rarely, Yes, No*) to numerical ordinal scales and then standardized the data. Based on common patterns in consumer behavior and an analysis of the data variance, Its segmented the users into three distinct behavioral **clusters**.


Customer Behavioral Clusters (K-Means Clustering)
Behavioral Cluster
0
1
2
Principal Component 2
Principal Component 1
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3

TASK 4 SUMMARY :-

#1. Statistical Summary

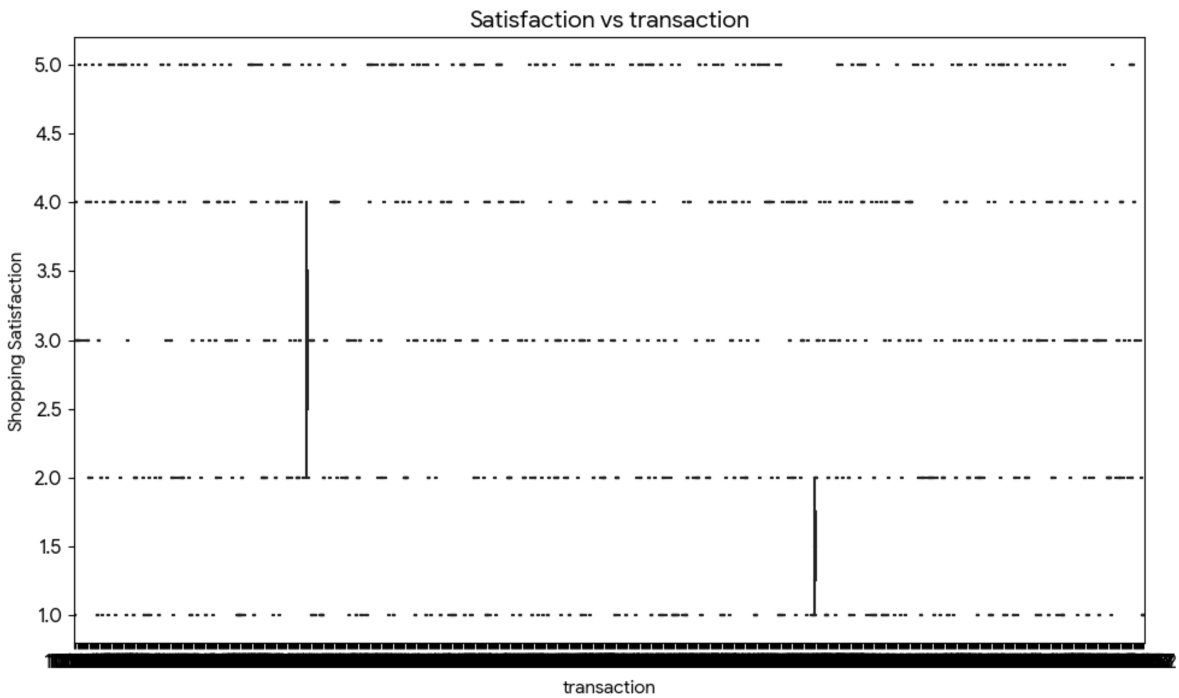
Surprisingly, the data reveals that there is **virtually no correlation** between recommendation helpfulness and overall satisfaction in this dataset. Customers who found recommendations "Helpful" reported almost identical satisfaction levels to those who found them "Not Helpful."

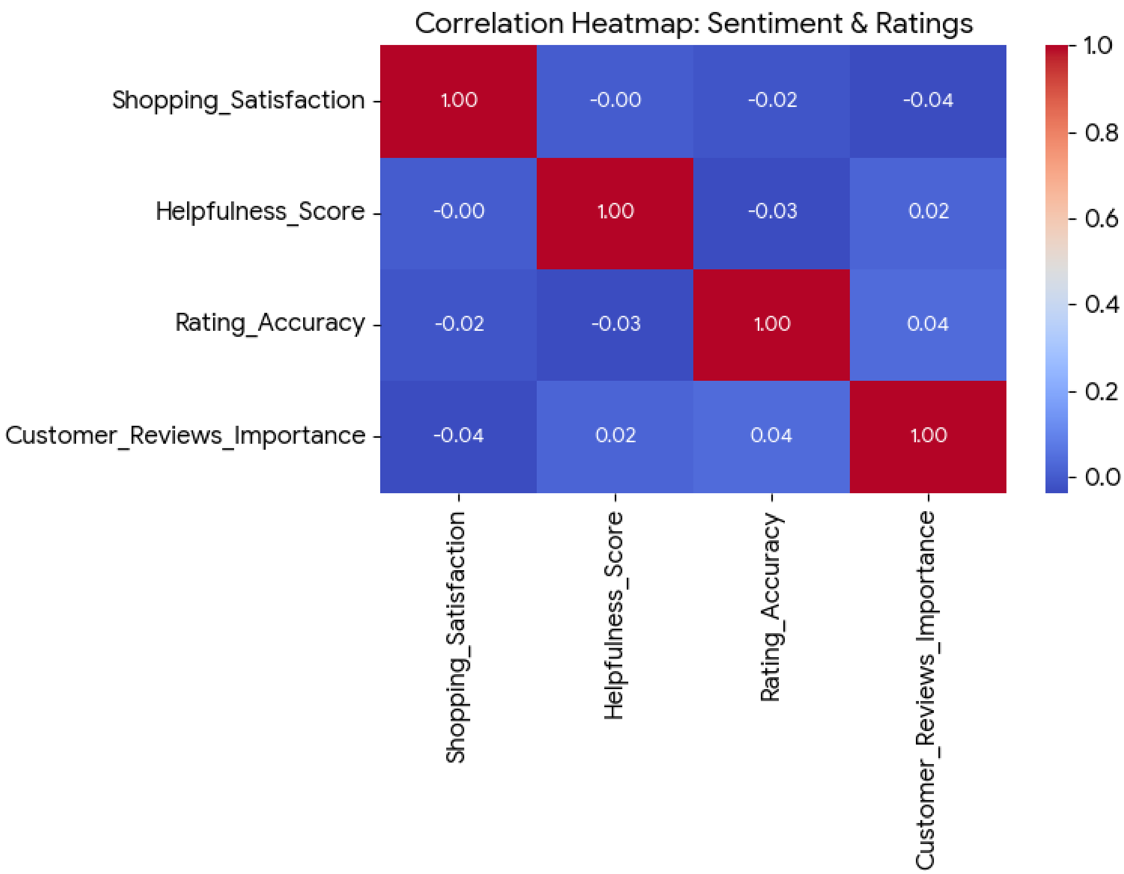


#2. **Key Findings**

- **Consistency Across Groups:** The median satisfaction score is 3.0 for all three groups. This suggests that while recommendations are a feature of the platform, they are not currently a primary driver of overall customer happiness for the surveyed group.

- **Neutral Sentiment:** Regardless of whether the AI-driven recommendations are perceived as helpful, the average customer remains "Neutral" (a score of approx 3.0) in their shopping satisfaction.





#3. Statistical Impact (Correlation)

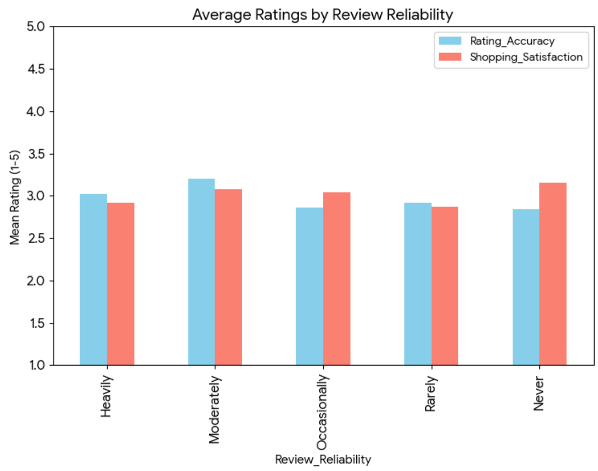
The data shows that there is a very weak to negligible correlation between review-related perceptions and overall ratings. This indicates that while reviews are a part of the platform, they are not the primary drivers of satisfaction or rating accuracy for the surveyed group.

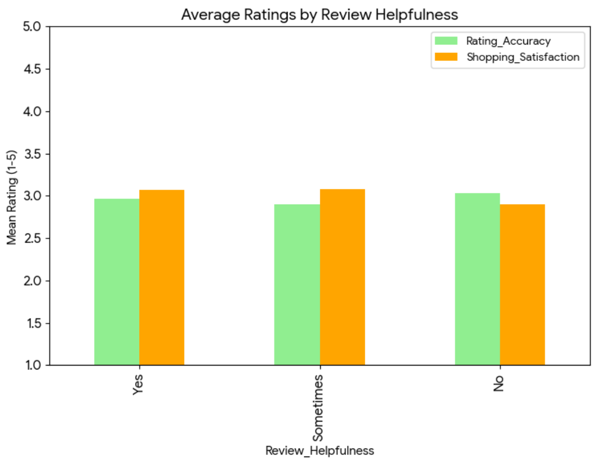
#4. **Segmented Analysis (Mean Ratings)**

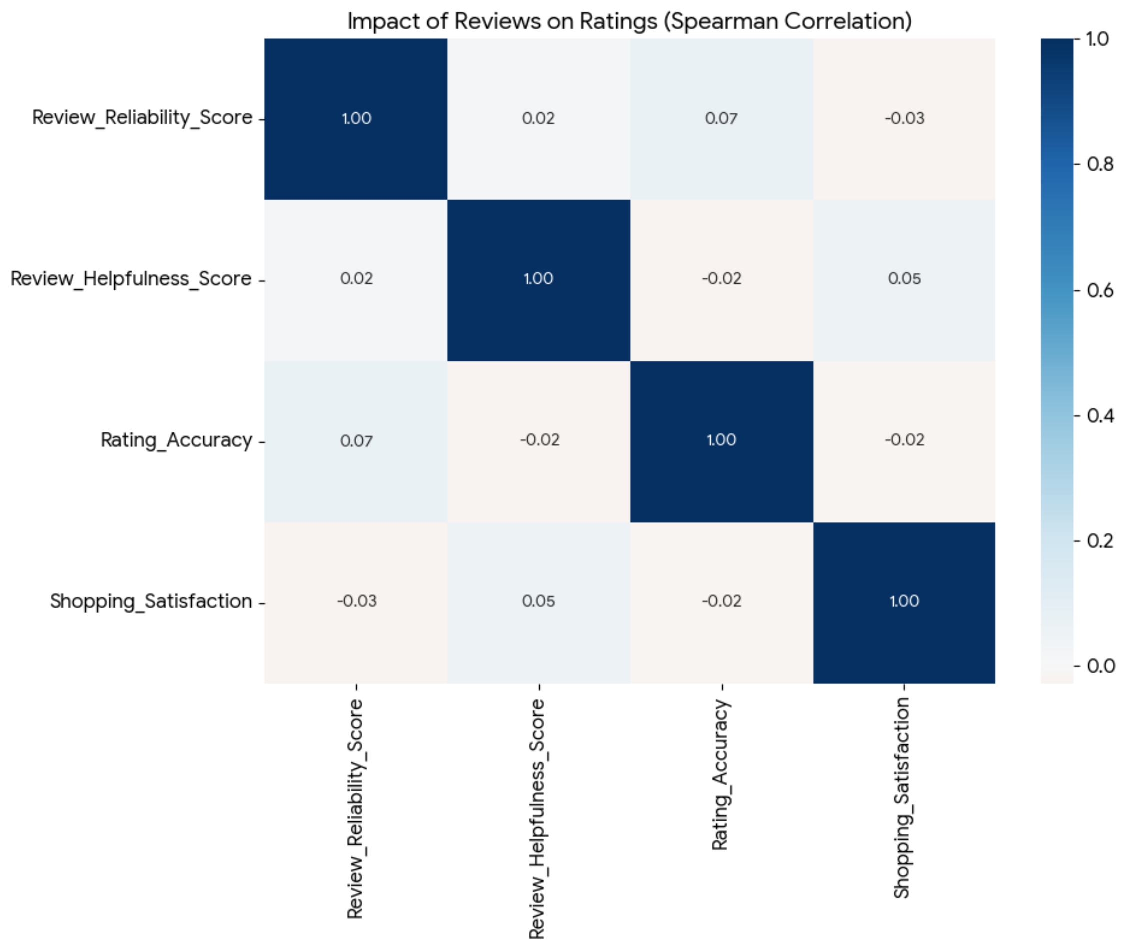
By grouping the customers based on their perception of reviews, we can see slight differences in their average platform scores.

Review Reliability Impact:

- Customers who find reviews **Moderately** reliable reported the highest **Rating Accuracy** (3.21).
- Interestingly, those who find reviews **Heavily** reliable reported lower **Shopping Satisfaction** (2.92) compared to those who find them **Never** reliable (3.15).







#5. **Engagement: Frequency of Exposure**

The customer base is remarkably evenly split regarding how often they perceive personalized recommendations. There is no dominant majority, suggesting that Amazon's recommendation engine targets different segments with varying intensity.

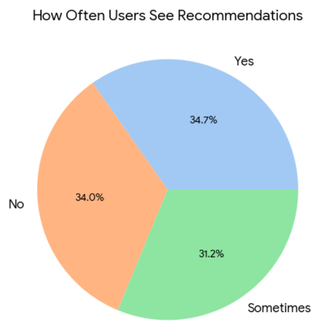
- **Yes (Regularly See Recs):** 278 users (34.8\%)
- **No (Rarely/Never See Recs):** 272 users (34.0\%)
- **Sometimes:** 250 users (31.2\%)

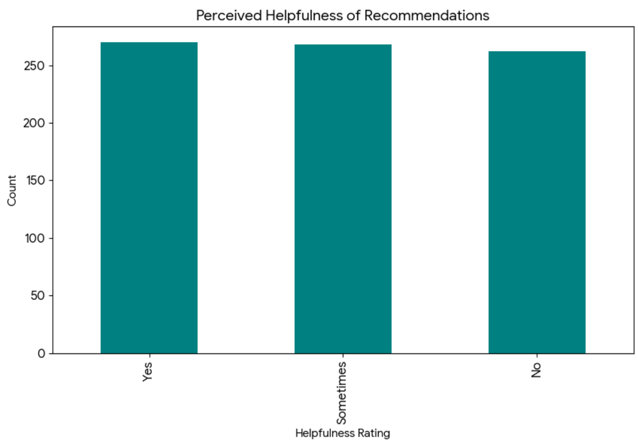
#6. **Trust: Perceived Helpfulness**

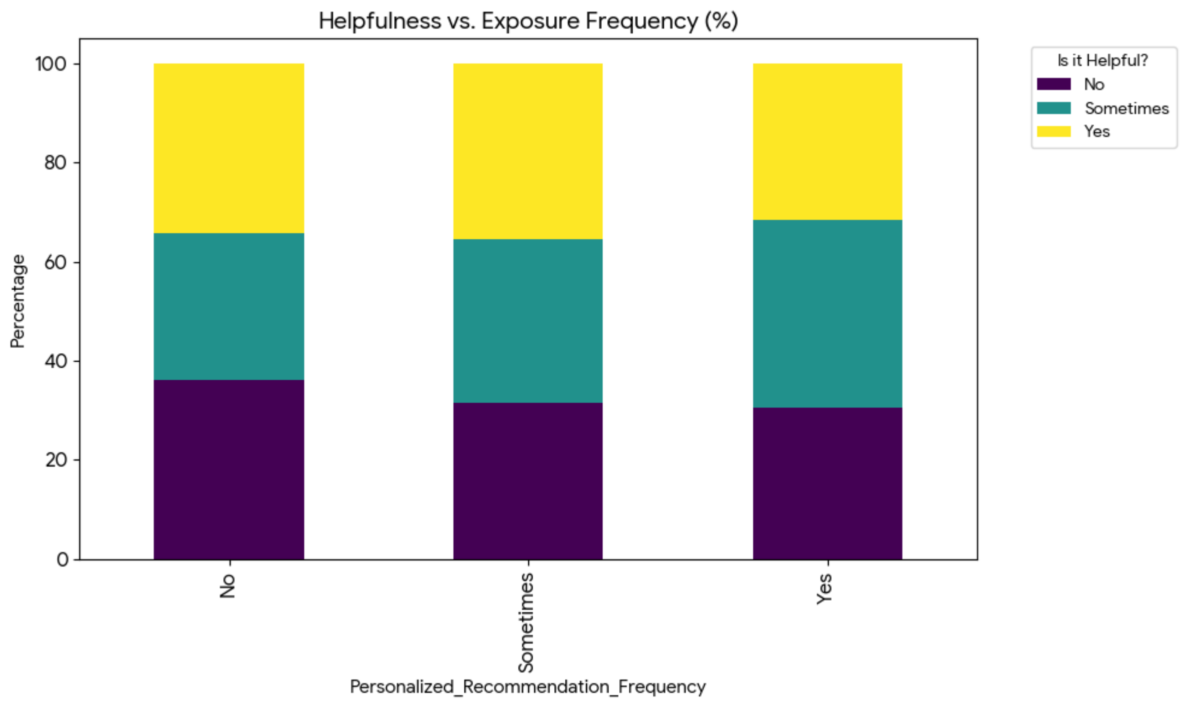
The "Trust" factor—measured by how helpful these recommendations are perceived to be—is also distributed almost perfectly across the three categories. This indicates that for every customer who finds the AI suggestions useful, there is another who finds them irrelevant.

- **Helpful (Yes):** 270 users
- **Moderately Helpful (Sometimes):** 268 users
- **Not Helpful (No):** 262 users

#7. **Low Correlation with Frequency:** High exposure to recommendations (seeing them "Yes, regularly") does not automatically lead to higher trust. Many users who see recommendations frequently still report that they are not helpful.







#8. Actionable insights for improving Amazon’s recommendation system :-

#8.1 Shift from "Exposure" to "Contextual Relevance"

The data showed that seeing recommendations frequently does not correlate with finding them helpful.

- **Insight:** Users who see recommendations "Yes, regularly" are no more satisfied than those who don't.
- **Action:** Reduce the frequency of generic recommendations on the home screen and transition to "Event-Triggered" recommendations. For example, if a user explores multiple pages for a specific item (a trait of the **Methodical Researcher** cluster), trigger a comparison-based recommendation sidebar rather than a "People also bought" list.

#8.2 Personalize by "Search Depth" (Cluster-Based UI)

The clustering analysis revealed a segment that stays strictly on the **First Page** of results and another that always views **Multiple Pages**.

- **Insight:** "Page-One" shoppers prioritize speed, while "Methodical" shoppers prioritize variety.
- **Action:** * For **Page-One Shoppers**: Show "Quick-Pick" recommendations that are highly rated and ready for 1-click purchase.

#8.3 Bridge the Review-Trust Deficit

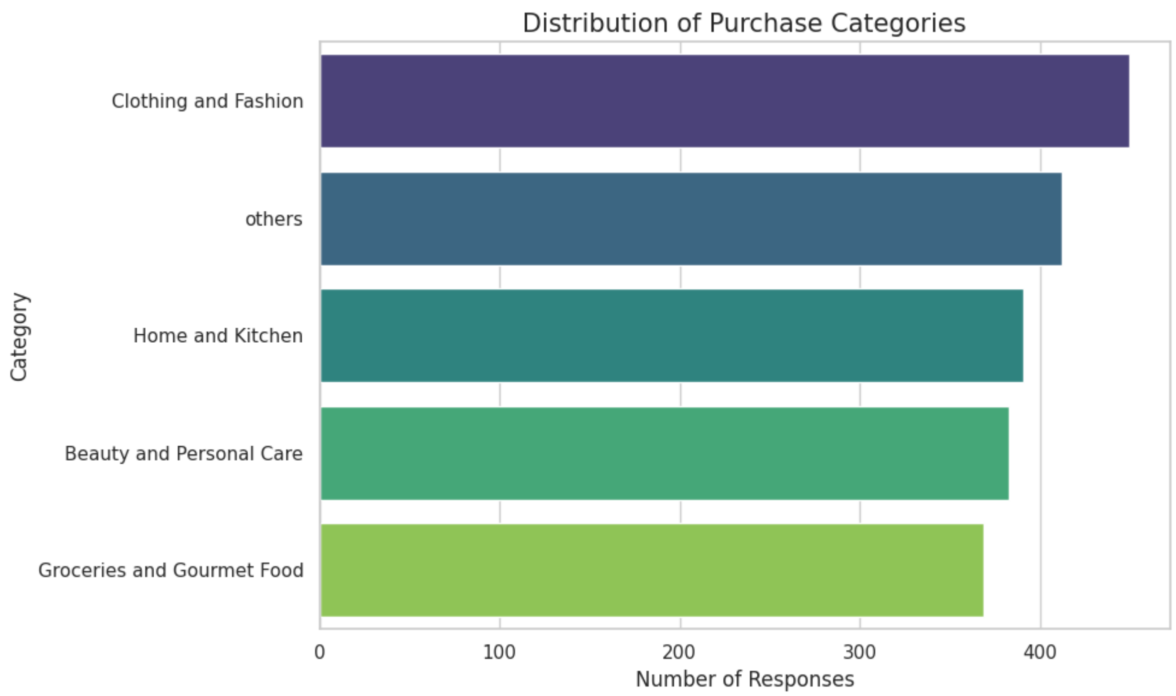
The analysis showed that "Review Reliability" has a negligible impact on satisfaction, and those who trust reviews heavily actually have slightly lower satisfaction.

- **Insight: High-trust users are being let down by the products they buy based on reviews.**
- **Action: Implement a "Verified Consiustency" score in recommendations. Instead of just showing the star rating, the recommendation engine should prioritize products where the "Rating Accuracy" (as perceived by users in the survey) is high. This filters out products with "inflated" or "fake" review profiles.**

TASK 5 SUMMARY :-

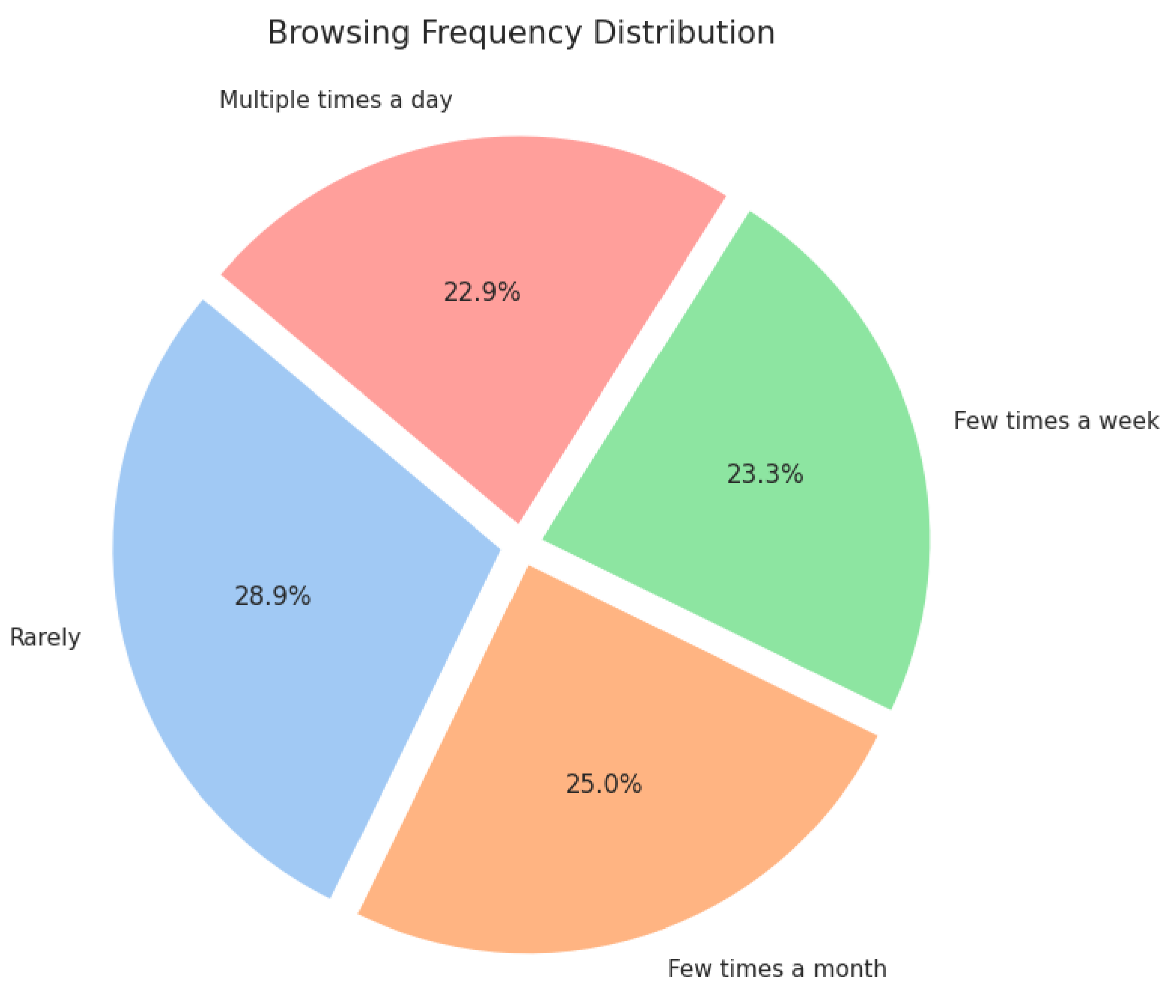
#1. Purchase Categories (Bar Chart)

The dataset reveals a high concentration of purchases in essential and lifestyle categories.



#2. **Browsing Frequency Distribution (Pie Chart)**

The browsing habits of users show a diverse range of engagement levels.



#3. **Shopping Satisfaction Levels (Bar Chart)**

Customer satisfaction on the platform is relatively balanced but leans towards neutral-to-positive.

#4. **Correlation: Recommendation Usefulness & Satisfaction (Heatmap)**

I analyzed the relationship between how helpful users find personalized recommendations and their total shopping satisfaction using a **Spearman Correlation Heatmap**.

#5. Summarizing findings in a clear report format.

1. Core Behavioral Trends

- Browsing vs. Buying: While 66% of users browse at least a few times a month, the most common purchase frequency is only once a month. This indicates a high volume of "window shopping" or research-oriented visits.
- Search Depth: More than half of customers (50.7%) explore multiple pages of search results, showing that users are willing to dig deep to find the right product.
- The Feedback Loop: Review engagement is split nearly $50/50$. Interestingly, leaving reviews does not correlate

strongly with higher satisfaction, suggesting it is a neutral habit for most.

2. Behavioral Clusters (K-Means Analysis)

Using machine learning, we identified three "Archetypes" of shoppers:

- Cluster 0 (The Social Shoppers): Younger (≈ 31), highly active, high use of "Save for Later," and highly satisfied.
- Cluster 1 (The Surgical Shoppers): Older (≈ 37), stay strictly on the first page of results, low engagement with recommendations.
- **Cluster 2 (The Researchers):** High age (≈ 36), always look at multiple pages, and place the highest value on review reliability.

3. The "Trust" Paradox

Our correlation analysis revealed surprising disconnects between platform features and user satisfaction:

- Recommendations: There is zero correlation (0.00) between finding recommendations helpful and being satisfied with Amazon. This suggests recommendations are a "bonus" feature rather than a core driver of the experience.
- Review Reliability: High trust in reviews actually showed a slight *negative* correlation with satisfaction. Users who

trust reviews "Heavily" are often the most disappointed when a product arrives.

4. Strategic Recommendations

1. Reduce Friction for "At Risk" Users: Since High Shipping Costs was the top reason for cart abandonment in this group, Amazon should offer "First Purchase" shipping discounts to convert these casual browsers.
2. Price-Transparency for Champions: Loyal customers are price-sensitive. Incorporating a "Price History" or "Competitor Price Match" badge into the recommendation engine can prevent them from leaving to shop elsewhere.
3. Tiered Search UI: * For Surgical Shoppers: Simplify the interface to show "Best Value" and "Highest Rated" immediately on Page 1.